MATH 147 — Proof Practice

Lectures 7–12 Fall 2026

Attempt each problem before looking at the hints in lecture. Write complete proofs.

Lecture 7 — Convergence and Boundedness

Problem 1. Prove directly from the definition that

$$\lim_{n\to\infty} \frac{3n+1}{n+2} = 3.$$

Problem 2. Let (x_n) be a convergent sequence. Prove that (x_n) is bounded, that is, there exists $M > 0$ with $|x_n| \le M$ for all $n \in \mathbb{N}$.

Lecture 8 — Algebra of Limits

Problem 3. Suppose $x_n \to x$ and $y_n \to y$. Prove that $x_n y_n \to xy$ using the triangle inequality.

Problem 4. Let $x_n \to L$ with $L > 0$.

(a) Show that there exists $N \in \mathbb{N}$ such that $x_n > L/2$ for all $n \ge N$.

(b) Deduce that $\sqrt{x_n} \to \sqrt{L}$.

Lecture 9 — Monotone Sequences

Problem 5. Define $x_1 = 1$ and $x_{n+1} = \sqrt{2 + x_n}$ for $n \ge 1$. Prove that the recursive sequence (x_n) is increasing and bounded, and find its limit.

Problem 6. Let $a_n = \left(1 + \frac{1}{n}\right)^n$. Show that (a_n) is increasing and bounded above by 3, and conclude that it converges.

Lecture 10 — Subsequences

Problem 7. Prove that if every subsequence of (x_n) has a further subsequence converging to L, then $x_n \to L$.

Lecture 11 — Bolzano–Weierstrass

Problem 8. Use the Bolzano–Weierstrass theorem to show that every bounded sequence (x_n) has a convergent subsequence whose limit lies in the interval $[\inf_n x_n, \sup_n x_n]$.

Lecture 12 — Cauchy Sequences

Problem 9. Suppose $|x_{n+1} - x_n| \le 2^{-n}$ for all $n \in \mathbb{N}$. Prove that (x_n) is a Cauchy sequence.

Problem 10. For bounded sequences (x_n) and (y_n), prove that

$$\limsup_{n\to\infty}(x_n + y_n) \le \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n.$$